Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: SolarSmart_Design_Challenge

Variant: New simulation variant

Tables on a building

System power: 32.7 kWp

PSG_iTech - India

Author

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Project summary

Geographical Site
PSG_iTech
India

Situation

Latitude	11.07 °N
Longitude	77.09 °E
Altitude	374 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
PSG_iTech
Meteonorm 8.1 (1991-2010), Sat=100% - Synthetic

System summary

Grid-Connected System

Tables on a building

PV Field Orientation
Fixed plane
Tilt/Azimuth 11 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Daily profile
Constant over the year
Average 220 kWh/Day

System information

PV Array

Nb. of modules	60 units
Pnom total	32.7 kWp

Inverters

Nb. of units	1 unit
Pnom total	25.00 kWac
Pnom ratio	1.308

Results summary

Produced Energy	51254 kWh/year	Specific production	1567 kWh/kWp/year	Perf. Ratio PR	81.26 %
Used Energy	80299 kWh/year			Solar Fraction SF	37.09 %

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General parameters

Grid-Connected System

Tables on a building

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	11 / 0 °

Sheds configuration	
Nb. of sheds	9 units
Sizes	
Sheds spacing	4.72 m
Collector width	4.11 m
Ground Cov. Ratio (GCR)	87.0 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Daily profile
Constant over the year
Average 220 kWh/Day

Hourly load	0 h	1 h	2 h	3 h	4 h	5 h	6 h	7 h	8 h	9 h	10 h	11 h	
	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	6.29	9.43	9.43	9.43	kW
	12 h	13 h	14 h	15 h	16 h	17 h	18 h	19 h	20 h	21 h	22 h	23 h	
	9.43	9.43	9.43	9.43	9.43	12.57	22.00	22.00	22.00	22.00	3.77	3.77	kW

PV Array Characteristics

PV module

Manufacturer	Waaree Energies
Model	WSMD-545
(Original PVsyst database)	
Unit Nom. Power	545 Wp
Number of PV modules	60 units
Nominal (STC)	32.7 kWp
Modules	4 string x 15 In series
At operating cond. (50°C)	
Pmpp	29.90 kWp
U mpp	563 V
I mpp	53 A

Inverter

Manufacturer	Polycab
Model	PSIT-25K-SM4
(Custom parameters definition)	
Unit Nom. Power	25.0 kWac
Number of inverters	2 * MPPT 50% 1 unit
Total power	25.0 kWac
Operating voltage	160-1000 V
Pnom ratio (DC:AC)	1.31
No power sharing between MPPTs	

Total PV power

Nominal (STC)	33 kWp
Total	60 modules
Module area	154 m²

Total inverter power

Total power	25 kWac
Number of inverters	1 unit
Pnom ratio	1.31

Array losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 176 mΩ
Loss Fraction 1.5 % at STC

LID - Light Induced Degradation
Loss Fraction 1.5 %

Module Quality Loss
Loss Fraction 0.5 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

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Array losses

IAM loss factor

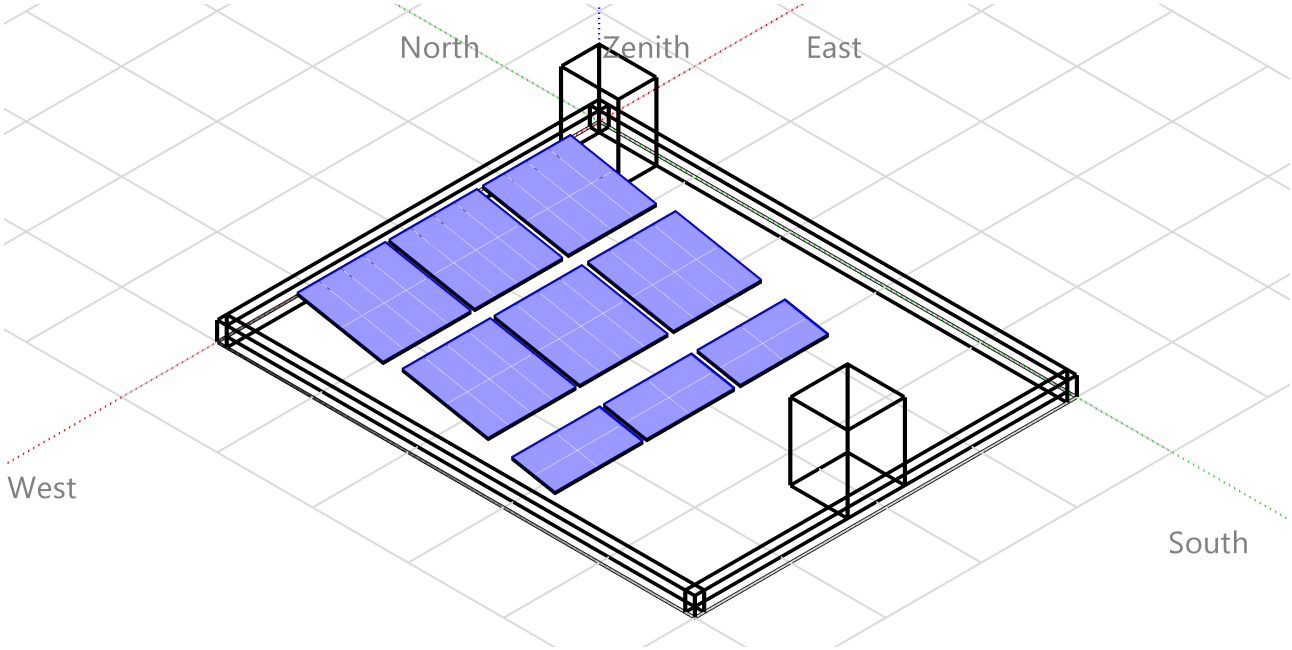
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

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Near shadings parameter

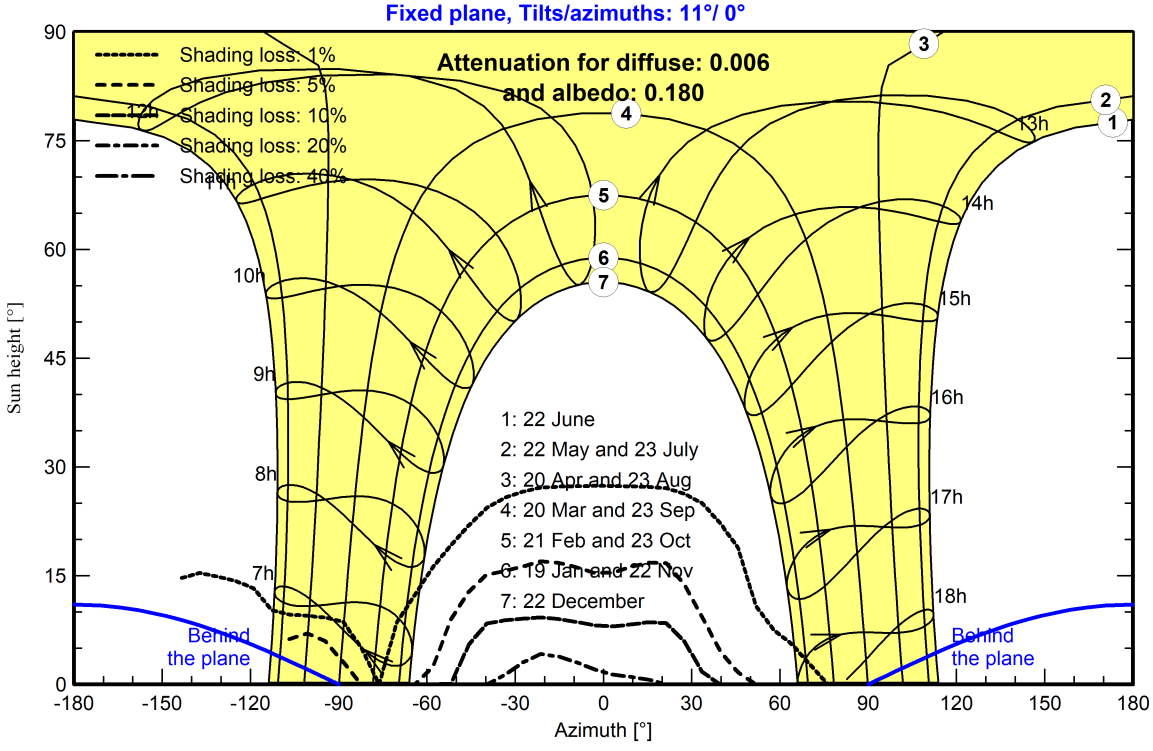
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 11° / 0°



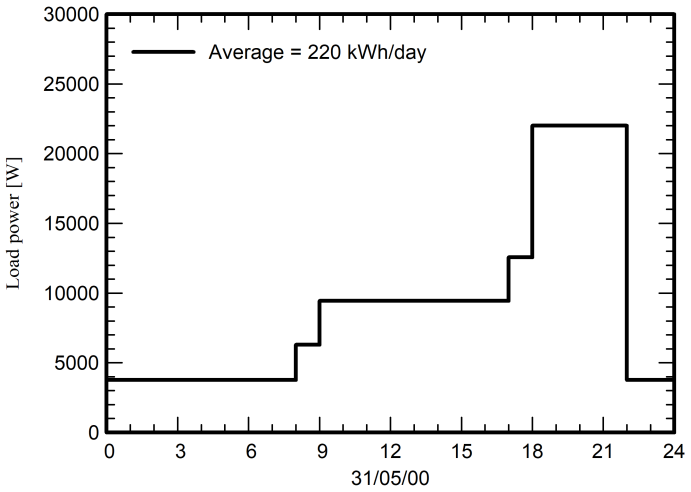
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Detailed User's needs

Daily profile, Constant over the year, average = 220 kWh/day

Hourly load	0 h	1 h	2 h	3 h	4 h	5 h	6 h	7 h	8 h	9 h	10 h	11 h	
	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	6.29	9.43	9.43	9.43	kW
	12 h	13 h	14 h	15 h	16 h	17 h	18 h	19 h	20 h	21 h	22 h	23 h	
	9.43	9.43	9.43	9.43	9.43	12.57	22.00	22.00	22.00	22.00	3.77	3.77	kW

Daily profile



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Main results

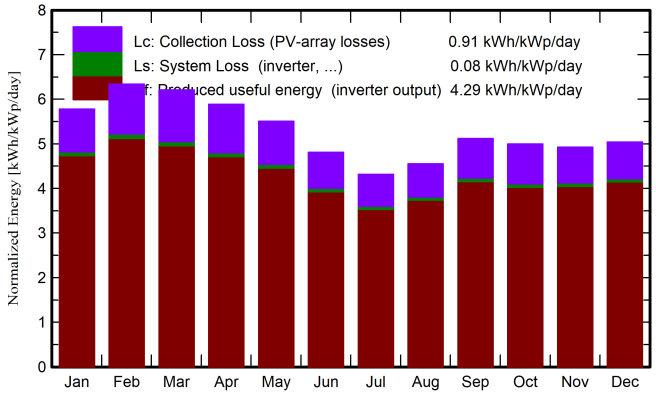
System Production

Produced Energy	51254 kWh/year	Specific production	1567 kWh/kWp/year
Used Energy	80299 kWh/year	Perf. Ratio PR	81.26 %
		Solar Fraction SF	37.09 %

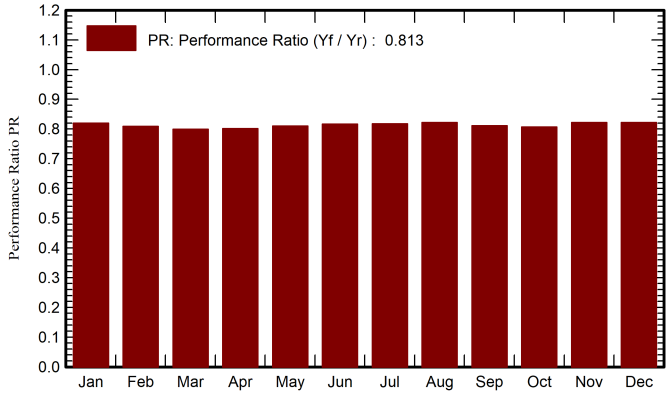
Economic evaluation

Investment		Yearly cost		LCOE	
Global	1,172,664.00 INR	Annuities	0.00 INR/yr	Energy cost	2.42 INR/kWh
Specific	35.9 INR/Wp	Run. costs	19,571.97 INR/yr		
		Payback period	4.9 years		

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

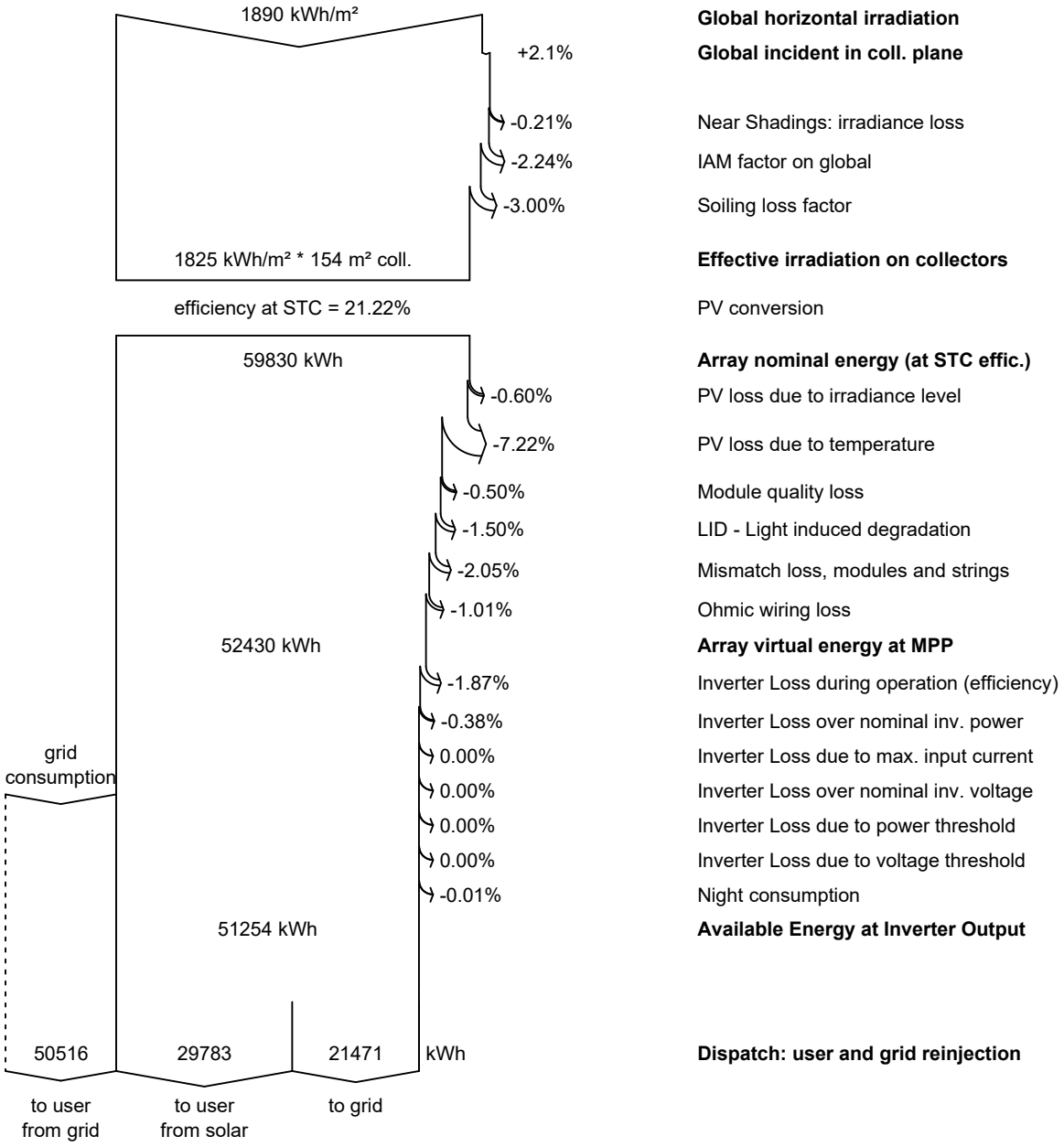
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	161.7	63.42	24.80	179.2	169.7	4891	6820	2636	2165	4183
February	165.5	62.81	26.54	177.5	168.6	4784	6160	2466	2231	3694
March	187.7	76.16	28.64	192.5	182.7	5127	6820	2687	2346	4133
April	179.4	79.91	29.01	176.7	167.4	4717	6600	2552	2076	4048
May	178.4	87.82	28.53	170.8	161.5	4612	6820	2652	1873	4168
June	152.5	78.05	26.46	144.3	135.9	3929	6600	2461	1393	4139
July	140.3	83.76	26.31	133.9	126.0	3654	6820	2387	1196	4433
August	144.8	91.32	26.01	141.1	133.0	3865	6820	2520	1271	4300
September	152.3	75.32	26.19	153.7	145.4	4158	6600	2322	1756	4278
October	148.9	71.39	26.20	154.9	146.7	4166	6820	2376	1711	4444
November	136.9	68.01	24.97	147.9	140.2	4050	6600	2276	1696	4324
December	141.7	63.11	24.47	156.4	148.2	4285	6820	2449	1757	4371
Year	1890.0	901.07	26.51	1928.9	1825.3	52237	80299	29783	21471	50516

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	E_Grid	Energy injected into grid
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

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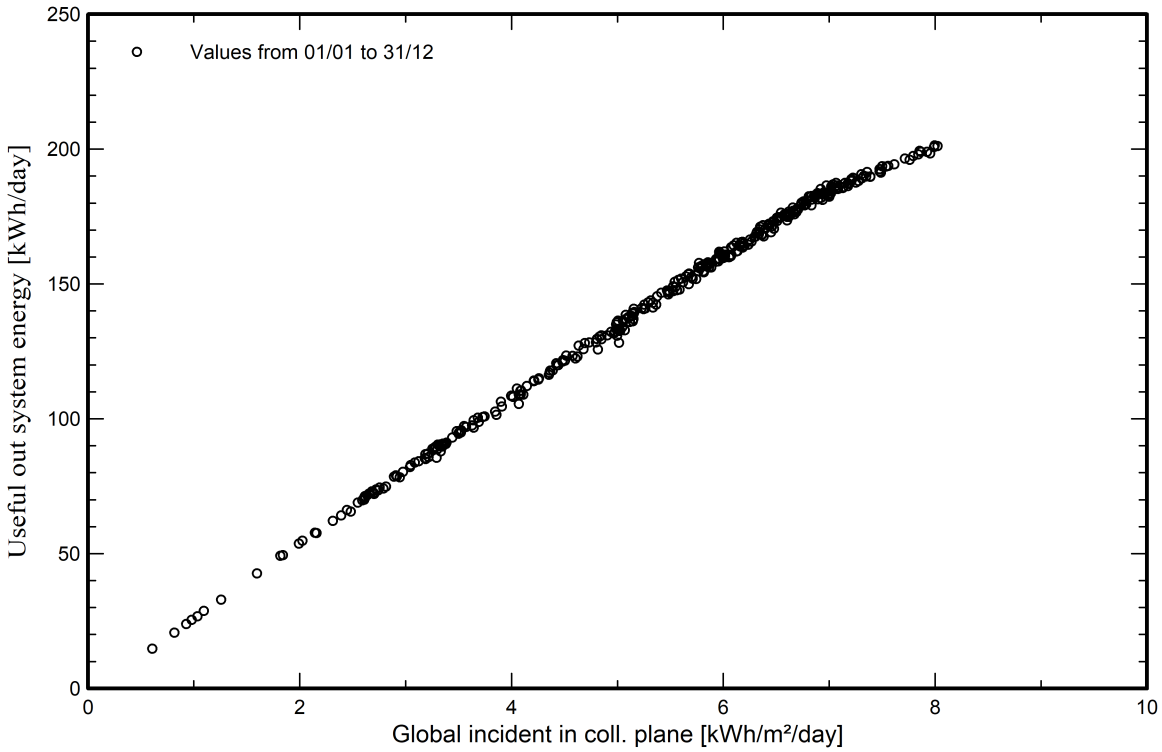
Loss diagram



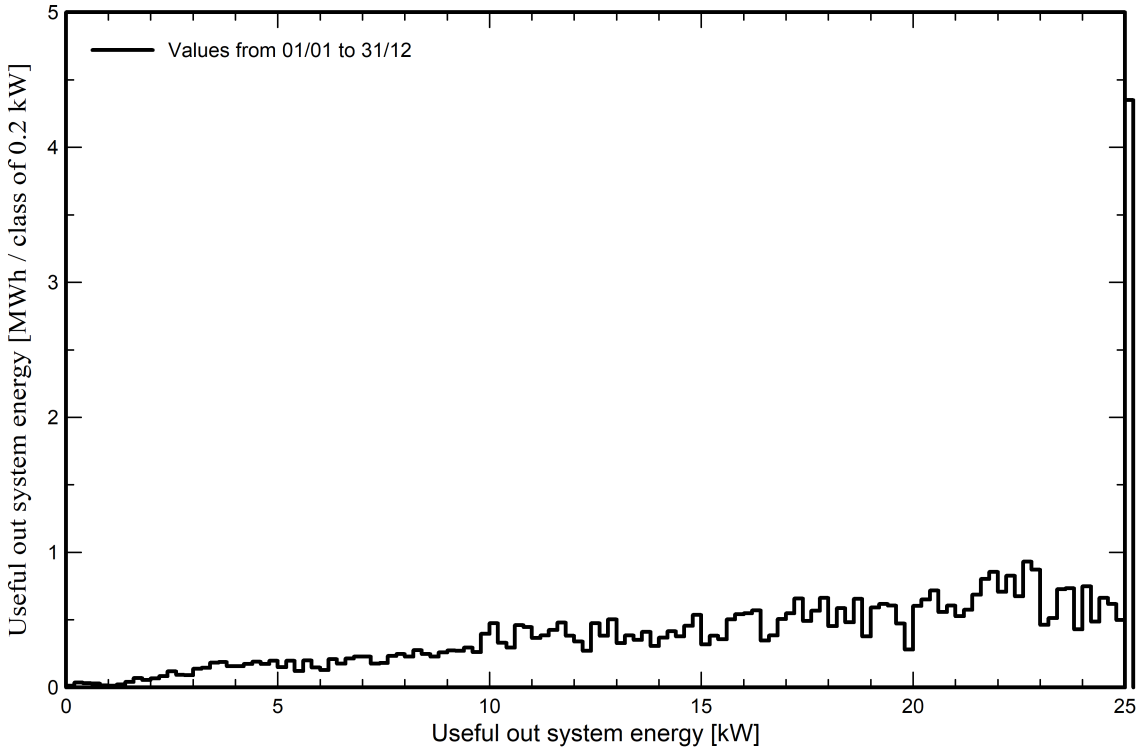
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



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P50 - P90 evaluation

Weather data

Source	Meteonorm 8.1 (1991-2010), Sat=100%
Kind	Monthly averages
Synthetic - Multi-year average	
Year-to-year variability(Variance)	6.2 %

Specified Deviation

Climate change	0.0 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

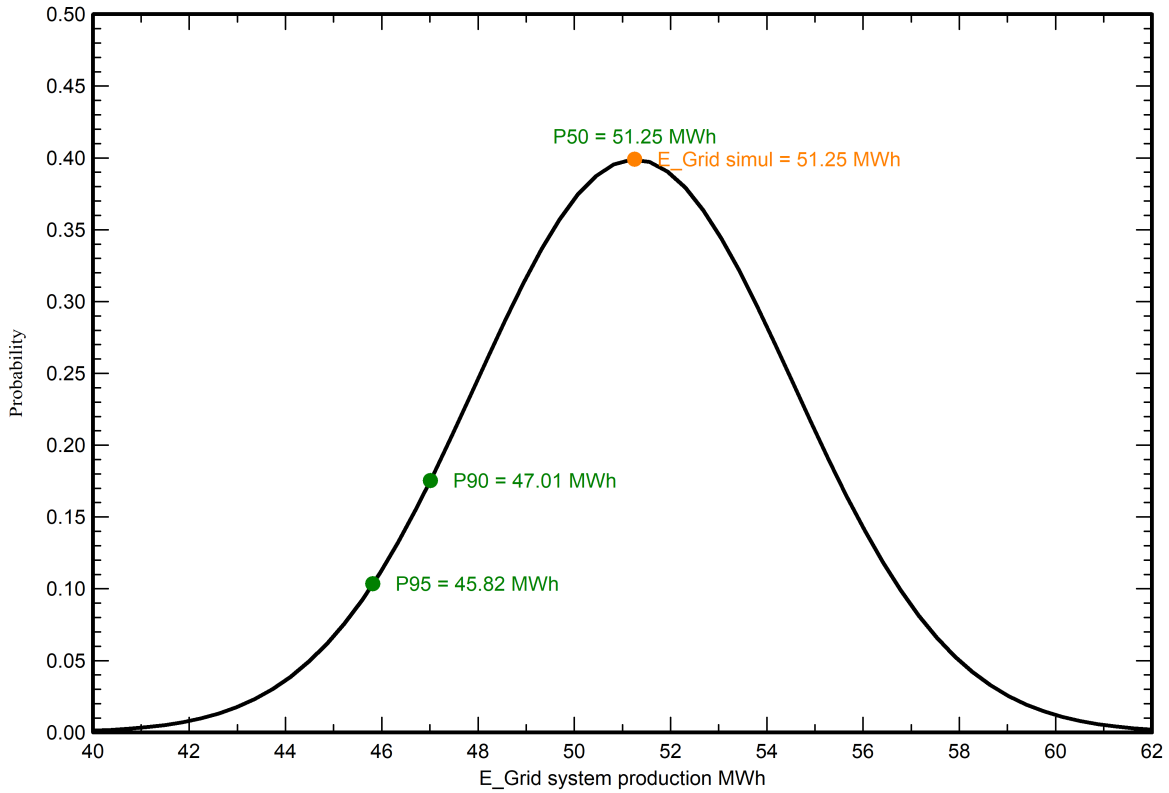
Global variability (weather data + system)

Variability (Quadratic sum)	6.5 %

Annual production probability

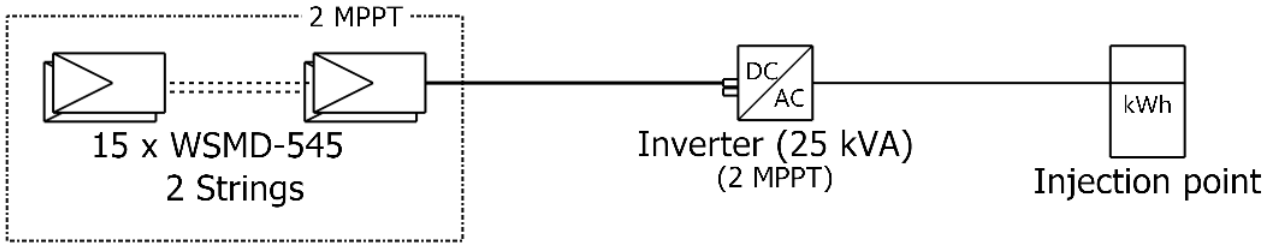
Variability	3.31 MWh
P50	51.25 MWh
P90	47.01 MWh
P95	45.82 MWh

Probability distribution



Single-line diagram

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PV module	WSMD-545
Inverter	PSIT-25K-SM4
String	15 x WSMD-545

	SolarSmart_Design_Challenge	
	VC0 : New simulation variant	07/10/26

Cost of the system

Installation costs

Item	Quantity units	Cost INR	Total INR
PV modules			
WSMD-545	60	10,900.00	654,000.00
Inverters			
PSIT-25K-SM4	1	137,500.00	137,500.00
Other components			
Mounting	9	16,666.67	150,000.00
Accessories, fasteners	9	5,555.56	50,000.00
Wiring	1	70,000.00	70,000.00
Surge arrester	1	24,300.00	24,300.00
Installation			
Global installation cost per module	60	872.00	52,320.00
Global installation cost per inverter	1	11,000.00	11,000.00
Transport + Grid Setup	1	23,544.00	23,544.00
		Total	1,172,664.00
		Depreciable asset	841,500.00

Operating costs

Item	Total INR/year
Maintenance	14,071.97
Maintenance	
Provision for inverter replacement	5,500.00
Total (OPEX)	19,571.97

System summary

Total installation cost	1,172,664.00	INR
Operating costs	19,571.97	INR/year
Useful energy from solar	29.8	MWh/year
Energy sold to the grid	21.5	MWh/year
Cost of produced energy (LCOE)	2.4224	INR/kWh

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Financial analysis

Simulation period

Project lifetime	25 years	Start year	2027

Income variation over time

Inflation	0.00 %/year
Production variation (aging)	0.50 %/year
Discount rate	8.00 %/year

Income dependent expenses

Income tax rate	0.00 %/year
Other income tax	0.00 %/year
Dividends	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (INR)	Depreciable (INR)
PV modules				
WSMD-545	Straight-line	25	0.00	654,000.00
Inverters				
PSIT-25K-SM4	Straight-line	25	0.00	137,500.00
Accessories, fasteners	Straight-line	25	0.00	50,000.00
		Total	0.00	841,500.00

Financing

Own funds	1,172,664.00 INR

Electricity sale

Feed-in tariff	3.00000 INR/kWh
Duration of tariff warranty	25 years
Annual connection tax	0.00 INR/kWh
Annual tariff variation	0.0 %/year
Feed-in tariff decrease after warranty	0.00 %

Self-consumption

Consumption tariff	8.50000 INR/kWh
Tariff evolution	0.0 %/year

Return on investment

Payback period	4.9 years
Net present value (NPV)	2,152,437.99 INR
Internal rate of return (IRR)	25.86 %
Return on investment (ROI)	183.6 %

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Financial analysis

Detailed economic results (INR)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Self-cons. saving	Cumul. profit	% amorti.
0	0	1,172,664	0	0	0	0	0	0	-1,172,664	0.0%
1	64,425	0	19,572	33,660	11,193	0	44,853	253,158	-896,728	23.5%
2	64,747	0	19,572	33,660	11,515	0	45,175	254,424	-639,870	45.4%
3	65,071	0	19,572	33,660	11,839	0	45,499	255,696	-400,772	65.8%
4	65,396	0	19,572	33,660	12,164	0	45,824	256,974	-178,206	84.8%
5	65,723	0	19,572	33,660	12,491	0	46,151	258,259	28,971	102.5%
6	66,052	0	19,572	33,660	12,820	0	46,480	259,550	221,822	118.9%
7	66,382	0	19,572	33,660	13,150	0	46,810	260,848	401,338	134.2%
8	66,714	0	19,572	33,660	13,482	0	47,142	262,152	568,440	148.5%
9	67,048	0	19,572	33,660	13,816	0	47,476	263,463	723,987	161.7%
10	67,383	0	19,572	33,660	14,151	0	47,811	264,781	868,777	174.1%
11	67,720	0	19,572	33,660	14,488	0	48,148	266,104	1,003,555	185.6%
12	68,058	0	19,572	33,660	14,826	0	48,486	267,435	1,129,011	196.3%
13	68,399	0	19,572	33,660	15,167	0	48,827	268,772	1,245,792	206.2%
14	68,741	0	19,572	33,660	15,509	0	49,169	270,116	1,354,496	215.5%
15	69,084	0	19,572	33,660	15,852	0	49,512	271,467	1,455,682	224.1%
16	69,430	0	19,572	33,660	16,198	0	49,858	272,824	1,549,870	232.2%
17	69,777	0	19,572	33,660	16,545	0	50,205	274,188	1,637,543	239.6%
18	70,126	0	19,572	33,660	16,894	0	50,554	275,559	1,719,152	246.6%
19	70,477	0	19,572	33,660	17,245	0	50,905	276,937	1,795,117	253.1%
20	70,829	0	19,572	33,660	17,597	0	51,257	278,321	1,865,828	259.1%
21	71,183	0	19,572	33,660	17,951	0	51,611	279,713	1,931,647	264.7%
22	71,539	0	19,572	33,660	18,307	0	51,967	281,112	1,992,914	269.9%
23	71,897	0	19,572	33,660	18,665	0	52,325	282,517	2,049,942	274.8%
24	72,256	0	19,572	33,660	19,024	0	52,684	283,930	2,103,026	279.3%
25	72,617	0	19,572	33,660	19,385	0	53,045	285,349	2,152,438	283.6%
Total	**1,711,075**	**1,172,664**	**489,299**	**841,500**	**380,276**	**0**	**1,221,776**	**6,723,650**	**2,152,438**	**283.6%**

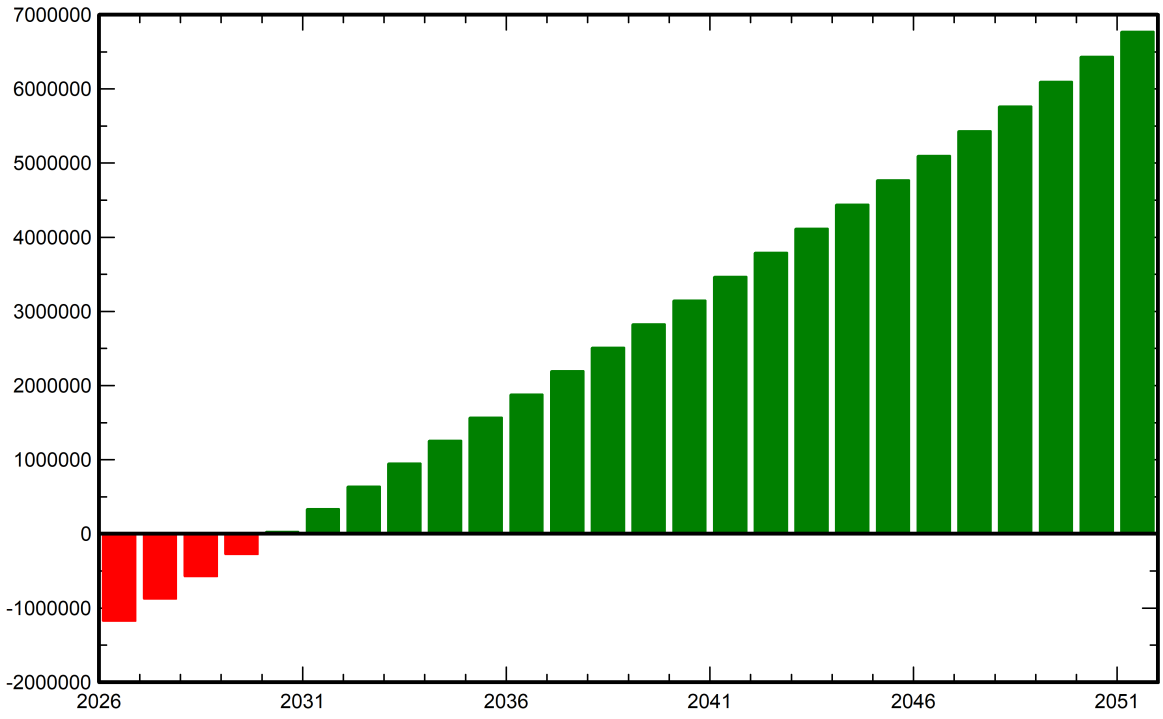
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Financial analysis

Yearly net profit (INR)



Cumulative cashflow (INR)



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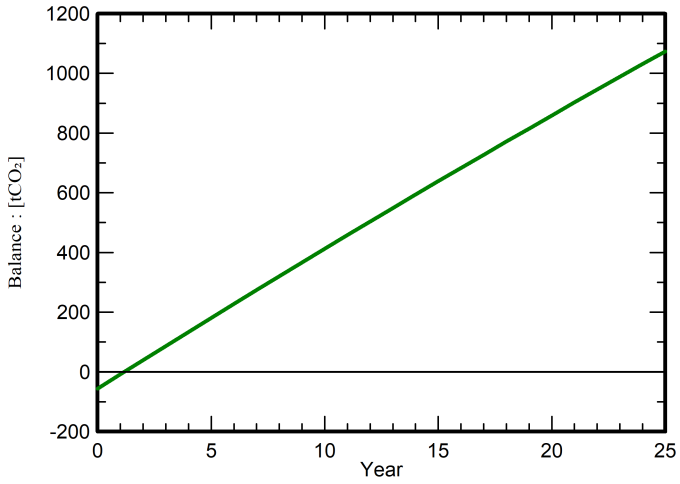
CO_2 Emission Balance

Total: 1073.8 tCO_2

Generated emissions
Total: 56.39 tCO_2
Source: Detailed calculation from table below

Replaced Emissions
Total: 1199.4 tCO_2
System production: 51.26 MWh/yr
Grid Lifecycle Emissions: 936 gCO_2/kWh
Source: IEA List
Country: India
Lifetime: 25 years
Annual degradation: 0.5 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	30.5 kWp	52272
Supports	6.24 kgCO2/kg	560 kg	3496
Inverters	619 kgCO2/units	1.00 units	619